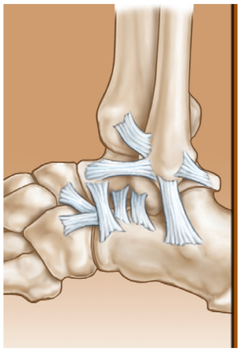


Sports Shoes and Orthoses

Nicholas LeCursi

SPORTS ORTHOSES

Foot orthoses—also commonly called orthotics, arch supports, and inserts—have been applied to the treatment of pathologic musculoskeletal conditions for more than a century.[1] The term *foot orthosis* refers to any orthotic device that is distal to the ankle. This naming convention was introduced in the early 1970s by the Committee on Prosthetic-Orthotic Education.[2]

A foot orthosis may be as simple as a prefabricated arch support available over the counter at the local pharmacy or as complex as a custom-designed device incorporating multiple supportive elements. Whatever the type, the beneficial effects of a foot orthosis are determined by its relative shape, stiffness, and compressibility with respect to the weight-bearing foot.

Although research suggests that foot orthoses are efficacious in the treatment of pathologic musculoskeletal conditions, outcome studies point to considerable variation in the reliability and effectiveness of orthotic treatment.[3] Clinical methodologies have been developed in an attempt to standardize treatment and improve the predictability of the clinical outcome.[4] Still, the results achieved by clinicians often differ, and interpractitioner variability appears to play a major role in the biomechanical response to orthotic intervention.[5,6] There is little objective evidence regarding *how* foot orthoses achieve favorable therapeutic results.[7] In spite of this, a growing body of objective evidence suggests that foot orthoses often produce positive clinical outcomes.[8-17] The potential benefits of a conservative, cost-effective treatment modality arguably justifies the continued prescription of foot orthoses and interest in orthotic-related research. An improved understanding of the mechanisms of action of foot orthoses is necessary to enhance clinical methodologies and better inform their prescription.

Research

Historically the focus of research has been the response of the lower extremity to the mechanical influence of an orthosis. Many studies have attempted to isolate the passive response of the limb by studying patients with neuropathy or utilizing cadaveric specimens.[18-20]

Study designs have attempted to isolate variables of interest and to measure the discrete mechanical influence of orthotic supportive elements such as base materials, interface materials, and metatarsal/scaphoid pads. Biomechanical studies suggest that foot orthoses are capable of altering plantar pressure distribution, foot and ankle kinetics/kinematics, and knee kinetics.[14,21-28]

Brodtkorb et al. found that the thickness of a metatarsal pad was more significant than its positioning in load shielding the metatarsal heads and that the maximum load-shielding effect occurred distal to the apex of the pad.[23]

Kogler et al. isolated the mechanical effects of arch supports as well as full-foot and forefoot wedges on plantar fascial strain. These cadaveric studies demonstrated the load-sharing effect of arch supports with the plantar fascia.[18,29,30] The medial column of the foot approximated the arch support's boundaries (Fig. 116.1).

Tsung et al. investigated the effect of arch supports on plantar pressure distribution in patients with sensory neuropathy. In this study, the method of casting determined the shape of the arch support for patients with neuropathy and strongly influenced the plantar pressure distribution. Foot shape was captured with the subjects in non–weight bearing, semi–weight bearing, and full weight-bearing conditions. In their study, the authors found that casting the foot in what they termed semi–weight bearing, with the subject standing with equal body weight on both feet, resulted in an orthotic shape with the least mean plantar pressure. The control of orthotic shape by the magnitude of the physiologic load produced orthoses with predictable, plantar pressure distributions.[28]

Guldemond et al. investigated the effects of metatarsal pads, arch supports, and full foot wedges on plantar pressure distribution in patients with neuropathy. The orthotic elements were added to a basic insole to alter its relative shape at the medial longitudinal arch and transverse arch.[20] This study revealed that the plantar pressure distribution could be altered in a predictable fashion and that the load-shielding effects of the various elements were additive (Fig. 116.2).

More recent studies suggest that in addition to their direct mechanical influence, foot orthoses can also evoke a neuromotor response that may play a significant role in their function.[31,32] Evidence suggests that tactile stimuli may, under certain conditions, affect foot and ankle posture.[25,26] Several authors have investigated the influence of comfort and surface texture on the biomechanical effectiveness of foot orthoses.[31,33,34]

Ritchie et al. found that plantar sensory feedback influenced midfoot pronation. The addition of mild, non–mechanically supportive surface features plantar to the medial arch elicited active arch elevation in walking subjects.[35]

Abstract

Research demonstrates that sports shoes and orthoses affect biomechanical variables, and that these devices can produce positive clinical outcomes. How those results are achieved, however, remains poorly understood. Comfort appears to play a key role in the delivery of orthotic care, sports shoe selection, and injury prevention. The clinician familiar with his or her patients' orthotic history and the many of sports shoe features that are available may help the athlete to achieve a comfortable fit, thus reducing the likelihood of sports injury.

Keywords

orthotic
sports shoes
comfort
injury prevention
supportive elements

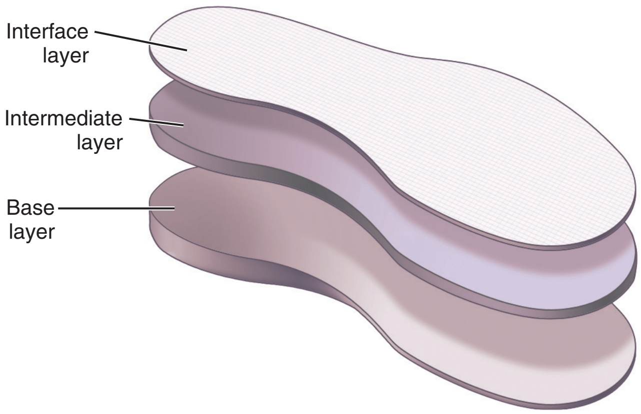


Fig. 116.1 Foot orthosis construction.

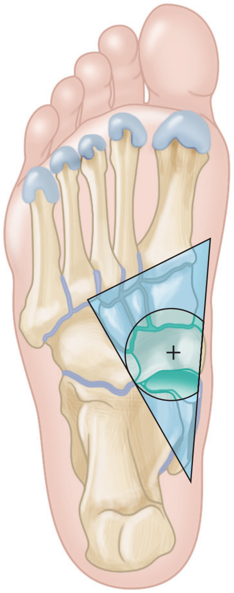


Fig. 116.2 Plantar aspect of the bones of the right foot showing primary support region for the longitudinal arch support mechanism of a foot orthosis. Partial circle indicates the position of the vertices of the orthotic support. (From Kogler GF, Solomonidis SE, Paul JP. Biomechanics of longitudinal arch support mechanisms in foot orthoses and their effect on plantar aponeurosis strain. *Clin Biomech [Bristol, Avon]*. 1996;11[5]:243–252.)

Mundermann et al. found that foot and ankle kinematics were related to orthotic comfort.[31] In these studies, orthotic use also significantly changed muscle activity.[31,36]

The evidence suggests that the neuromotor response to orthotic stimuli may play a significant role in the orthotic influence on biomechanical function.[32,33,37] Some manufacturers have even begun incorporating tactile elements in products based on this evidence. At this writing, the orthotic action upon a sensate, volitional foot appears to comprise a complex combination of direct mechanical action and the neuromotor response to the orthotic stimulus.[6,31,33,35]

Further exploration of the tactile response to orthotic mechanical action and the individual and combined mechanisms of action of functional elements in foot orthoses may help to enrich our understanding of the complex nature of foot orthosis design. An improved understanding of these functional elements may ultimately also enhance communication for clinical collaboration and research.

Clinical Applications

Injuries of the lower extremities are extremely common in runners,[38] and the majority of running injuries are associated with overuse.[39] Although training distance and history of previous injury are primary risk factors, a significant contributing factor is biomechanical abnormality of the lower extremity.[14,38] At this writing, there is evidence to suggest that foot orthoses may be efficacious in the treatment of metatarsalgia, plantar fasciitis, tendinopathies, stress fractures, compartment syndrome, and shin splints.[12-14,21,22,39-43] Foot orthoses may also be efficacious in the treatment of posterior tibial syndrome and foot pain secondary to postural pes planus and pes cavus.[9,10,44,45]

The complexity of the action of foot orthoses in treating foot pathology derives from the complexity of foot and ankle biomechanics and the neuromotor response to orthotic action. The clinician's initial approach to orthotic design is mechanically focused. The process of orthotic design typically begins with identification of the nature and severity of the patient's biomechanical deficits. Orthotic supportive elements are selected with the intent of providing a mechanical influence that may beneficially affect the biomechanical deficits.

The delivery of orthotic care is an iterative process of optimization of the orthotic design based on both subjective and objective clinical indicators. Patient comfort is an important clinical indicator utilized by most clinicians during orthotic fitting and optimization. Comfort in this sense implies not just the tactile feel of the orthosis but also the beneficial change in biomechanical variables associated with orthotic treatment of the pathology.[31] It is difficult to predict the response of a specific foot to orthotic support, which may account for some of the variability in patient satisfaction that occurs when the same condition is treated with a similar orthotic design. Despite having acquired an appropriately prescribed orthosis, many patients require multiple adjustments to achieve pain relief, and such adjustments are critical to ensuring the patient's satisfaction. Some basic guidelines that may serve as a starting point for the prescription of orthoses in the treatment of sports-related injuries are shown in Table 116.1. Note that these guidelines are intended to serve only as a starting point.

It is helpful to consider the focal influence of discrete orthotic supportive elements when foot orthoses are being prescribed to treat pathologic musculoskeletal conditions. There is some evidence, for example, suggesting that a simple flat insole added to a shoe may alter the comfort, pressure distribution, friction, or resistance to bacterial growth within the shoe.[24,46-49] Polyurethanes and elastomers such as Poron and Spenco, respectively, are examples of resilient insole materials that may enhance a shoe's comfort. Although cushioned insoles influence pressure distribution, it should be noted that there is evidence to suggest that they appear to have a questionable influence overall on shock attenuation.[50] Leather, vinyl, and silver-impregnated fabrics such as X-Static® are examples of interface layers that may influence friction and discourage bacterial growth.

TABLE 116.1 **Guideline for the Orthotic Treatment of Sports-Related Pathologies**

Pathology	Elements of Orthotic Support in Foot Orthoses
Sesamoiditis with no other postural abnormality	• Dancer pad—well out for the sesamoids. • P-cell foam insole with relief over first metatarsal head. • Addition of a reverse Morton extension may sometimes provide additional symptomatic relief. This stiffens the forefoot while minimizing the pressure over the sesamoids.
Metatarsalgia	• The height of the pad may be more effective than its placement in relieving metatarsal head pressure. • Having the pad placed immediately plantar to the metatarsal head may actually increase pain. • Additional use of a Morton extension may decrease pain secondary to elimination of the metatarsal pad extension.
Morton neuroma with no other postural abnormality	• Metatarsal pad added to native shoe insole with apex of pad at neuroma. • Metatarsal pad added to prefabricated orthosis with apex of pad at neuroma.
Plantar fasciitis with mild pes planus	• Consider relative height of orthosis arch support with respect to the weight-bearing foot. • Full support, with soft interface may be most effective at decreasing plantar fascia strain. • Addition of pressure relief with a cushioned heel for calcaneal pain may provide symptomatic relief.
Mild pes planus	• Three-quarter-length or full-length prefabricated insole with a supportive arch.
Moderate to severe pes planus	• Full-length orthotic with hindfoot inversion and medial arch support. • Consider planal dominance and nature of postural abnormality when selecting and balancing support for hindfoot, midfoot, and forefoot.
Pes cavus	• Soft custom orthosis may provide symptomatic relief by reducing plantar fascial strain. • In some cases it may be beneficial to provide full arch support where there is discomfort due to rigid pes cavus.
Diabetes	• Full-length accommodative orthotic fabricated from Plastozote.

The addition of discrete pads to the native shoe insole may enhance the support or comfort of a shoe. Metatarsal pads and bars, arch pads, lifts, and wedges are support elements that are available in various materials including felt, elastomers, and polyurethanes. Felt is typically supportive, whereas elastomers and polyurethanes are typically more resilient and conformable. Several examples of orthotic materials, pads, and their applications are shown in Table 116.2.

The decision whether to prescribe prefabricated or custom foot orthoses involves consideration of the posture of the weight-bearing foot. The inclusion of supportive elements in the design of the orthosis that are intended to treat biomechanical deficits should also be considered. Comfort may associate with the blending of the supportive elements with the anatomical contour, and this may best be achieved using custom devices.

The primary application of prefabricated foot orthoses is the enhancement of medial longitudinal arch support. The shape of these orthoses is designed to fit the average foot; therefore their application is most often limited to feet with normal arch height.

Custom foot orthoses fall into two categories, accommodative and corrective. The basis for the orthotic design of custom foot orthoses is an anatomic model produced by casting, foam impression, contact, or optical scan of the patient's foot.

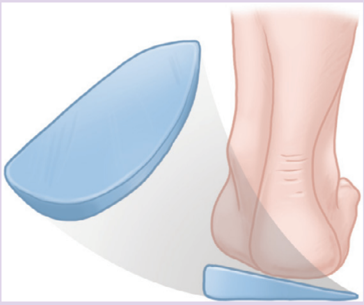
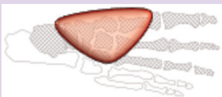
The shape of the patient's foot in weight bearing determines the shape of an accommodative foot orthosis. Most commonly these orthoses are designed to redistribute focal pressure or to minimize the mean plantar pressure. Rigid deformities of the foot such as posttraumatic pes planus and arthrosis from a Lisfranc injury are best treated with an accommodative orthotic, as any attempt to realign the foot will create increased focal pressure and pain. Diabetic foot orthoses are another example of an accommodative type.[51]

The shape of corrective foot orthoses is also anatomic, but with the inclusion of intentionally incongruous contours termed *modifications*. Examples of anatomic mold modifications used to create orthotic supportive elements are enhanced or reduced medial longitudinal arch supports, metatarsal pads, medial or lateral heel wedges, and medial or lateral forefoot wedges. The intent of these modifications is the seamless incorporation of supportive elements into the anatomic contour. These elements are often intended to influence foot and ankle kinetics and kinematics by redistributing plantar pressure. The Functional Foot Orthosis and the University of California Biomechanics Laboratory (UCBL) orthosis are two examples of corrective types.

Materials selection is an important aspect of foot orthosis design. Most foot orthoses comprise multiple layers, including a base and intermediate and interface layers. The use of multiple layers offers the combined benefits of their individual properties (Fig. 116.3). Firm base materials such as ethylene vinyl acetate (EVA), with compressibility like that of shoe outsoles, are typically used for the base layer of what are termed *soft foot orthoses*. Rigid base layers employ materials such as polypropylene, copolymer, and fiber glass or carbon lamination. Laminated foot orthoses typically offer the highest stiffness while taking up the least room in the shoe.

Soft, resilient, and conformable intermediate-layer materials include polyurethane foams, polyethylene foams, and elastomers. The intermediate layer helps to disperse plantar pressure and improve comfort.

TABLE 116.2 Common Foot Orthoses, Pads, and Materials

Component	Example	Intended Application	References
Foam-cushioned insole	Polyurethane, [a]Poron EVA, [a]P-cell polyethylene, [a]Plastozote	• Decrease impulse • Enhance comfort • Decrease focal plantar pressure	Mills et al.[46] Paton et al.[49] Chiu and Shiang[24]
Antifriction insole	Leather interface	• Decrease shear • Reduce callus formation (diabetic applications)	Yavuz et al.[77] Lu[48]
Antibacterial insole	Silver-impregnated fabric, [a]X-Static®	• Discourage bacterial/fungal growth	Sedov et al.[47]
Metatarsal pad		• Decrease metatarsal head plantar pressure	Brodtkorb et al.[23] Guldemond et al.[20]
Cushioned heel cup		• Decrease focal plantar heel pressure	Perhamre et al.[25]
Medial/lateral heel wedge		• Influence center of pressure at heel • Influence varus/valgus moment at heel • Influence rate of change of inversion/eversion at heel • Influence varus/valgus moment at knee	Huerta et al.[26,27] Leitch et al.[78] Bonanno et al.[79]
Full-foot medial/lateral wedge	Cork, EVA, or polyurethane Preformed wedge material 	• Influence varus/valgus moment at knee • Influence plantar fascial strain	Shelburne et al.[80] Kogler et al.[30]
Arch support (scaphoid pad)		• Influence plantar fascial strain • Influence plantar pressure distribution • Influence joint angles of the weight-bearing foot	Kogler et al.[18,29] Tsung et al.[28] Kitaoka et al.[19]

EVA, Ethylene vinyl acetate
[a]Acor Orthopaedics, Inc. Cleveland, OH; www.acore.com

The interface layer of a foot orthosis is typically leather, vinyl, or foam; it may enhance the durability and hygiene of the orthosis or alter the friction between the foot and the orthosis. Polyethylene foam such as Plastozote may enhance moisture control, friction, or pressure distribution of the interface layer. Fabrics may enhance the comfort, moisture control, or odor control of the orthosis.

One or more of the supportive elements may be incorporated into the orthotic design by casting or scanning, modification of the anatomic model, and materials selection for fabrication. The function of the orthosis is evaluated during its fitting and optimized for the patient's comfort.

SPORTS SHOES

Because early humans were largely dependent on hunting, one can postulate that the earliest footwear was used in running. With civilization's advancement and socialization, shoes took on symbolic functions.[52] Papyrus sandals for religious ceremonies and jeweled sandals for high-fashion gatherings have been

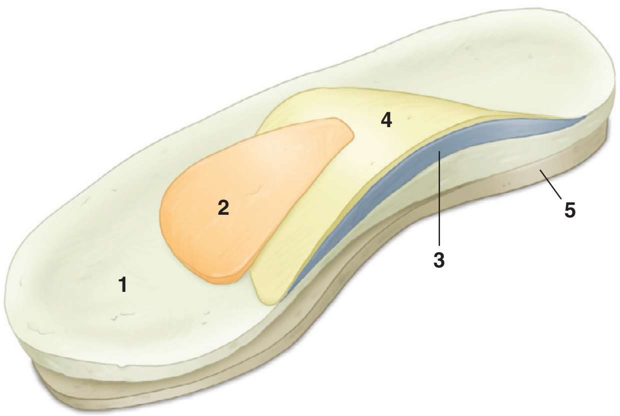


Fig. 116.3 Right angular front view of basic insole with components. *1,* Basic insole; *2,* metatarsal dome; *3,* "normal" arch support; *4,* "extra" arch support; and *5,* wedge (medial). (From Guldemond NA, Leffers P, Schaper NC, et al. The effects of insole configurations on forefoot plantar pressure and walking convenience in diabetic patients with neuropathic feet. *Clin Biomech [Bristol, Avon].* 2007;22[1]:81–87.)

discovered at the burial sites of Egyptian pharaohs.[53] Although these have little to do with sports shoes, they do foreshadow the current specialization and fashionable designs of athletic footwear. A shoe designed for sports alone did not come into existence until the latter half of the 19th century. The croquet sandal known as the *sneaker* had a fabric upper, rubber sole, and laces.[52,54,55] Further sports shoe development followed in the late 1800s with the demand for durable but lightweight shoes. It is from these developments that the multibillion-dollar athletic shoe industry evolved in an explosion of sport-specific footwear. Today's casual and professional athletes may select from a broad range of shoes designed for sprinting, running, cross training, volleyball, basketball, hiking, dance, and bowling, among other activities.

Over the last half century, significant progress has been made in the design, materials, and methods used to fabricate athletic shoes. Manufacturers have flooded the market with stylish models. These shoes are packed with inventive features ranging from mesh uppers to directionally siped soles. New features tend to fall in and out of fashion, however, suggesting that the consumer's selection of athletic shoes for sports activities may at times be more about form than function. In reality, the design and performance of the modern athletic shoe has dramatically improved. Manufacturers and independent researchers have performed studies to demonstrate the biomechanical influence of shoe features. While there is evidence that many of these features influence biomechanical variables, their efficacy in the prevention of sports injuries or in the enhancement of performance is less clearly understood. Although it remains difficult to identify specific aspects of design that will reduce sports injuries for a given athlete, fit and comfort appear to play a significant role in injury prevention.[56,57]

Clinicians have understandably been reluctant to recommend specific footwear for their patients. This reluctance is warranted given the almost overwhelming number of shoe models available, the pace of new product introduction, and the highly individualized nature of shoe fit and comfort. It is beneficial that consumers have a broad selection of shoe models from which to choose. To assist the clinician in helping to guide their patients in their selection, the next section describes some of the features designed into the modern sports shoe and their intended application.

Anatomy of the Sports Shoe

In simplest terms, the shoe can be broken down into the bottom and the upper. The bottom protects the foot, interfaces with the floor, and comprises the outsole, wedge, midsole, and insole. The upper covers the foot, supports the ankle, and comprises the toe box, tongue, closure, collar, and heel counter (Fig. 116.4). Each of these parts plays a role in the form, function, and fit of the shoe. Material properties play a major role in shoe performance, support, and durability.

Outsole

Advanced manufacturing methods have been developed to give modern athletic shoe sole materials unique physical properties. The primary function of the shoe outsole is to grip the running surface and resist wear. Shoe outsoles may be made of carbon rubber, the same material used for car tires. Carbon rubber is extremely durable, with exceptional wear resistance. Greater hardness imparts greater wear resistance and may distribute the contact forces from the outsole to the midsole more effectively. An outsole that is too hard, however, may impart a "stiff feel" to the shoe. Conversely, an outsole that is too soft may reduce the stability of the shoe.[58] Blown rubber is another example of an outsole material that is sometimes used for the forefoot section of a sports shoe. This material is more flexible and light in weight than carbon rubber but is less durable.

The coefficient of friction (COF) for shoe outsoles may range from 0.1 to 0.34 or higher but is heavily dependent upon the contact surface and is strongly influenced by moisture.[59,60] Research has shown that sole friction is time- and temperature-dependent as well; it also depends on tread design, even on hard flat surfaces.[60] Shoe manufacturers have employed outsole materials with friction properties suitable for specific sports applications. The nonmarring soles used for volleyball and high-friction outsoles for hiking are two examples.

Midsole

As stated previously, one function of the outsole is distribution of the compressive load to the midsole. Shoe midsole design has received much attention from shoe manufacturers. Because this is the primary dynamic element of the shoe sole, shoe designers have added ever more creative features to the midsole in an effort to improve stability, attenuate shock, and even enhance motor activation.

Shoe midsoles are most typically made of EVA, polyurethane, or some blend of thermoplastic elastomers (TPEs).[61] Advanced methods have been developed to vary the properties of midsole materials during manufacture. For example, manufacturing methods have been developed that facilitate the selective modification of midsole density along the sole contour, creating regions with different mechanical characteristics.[62] Research has shown

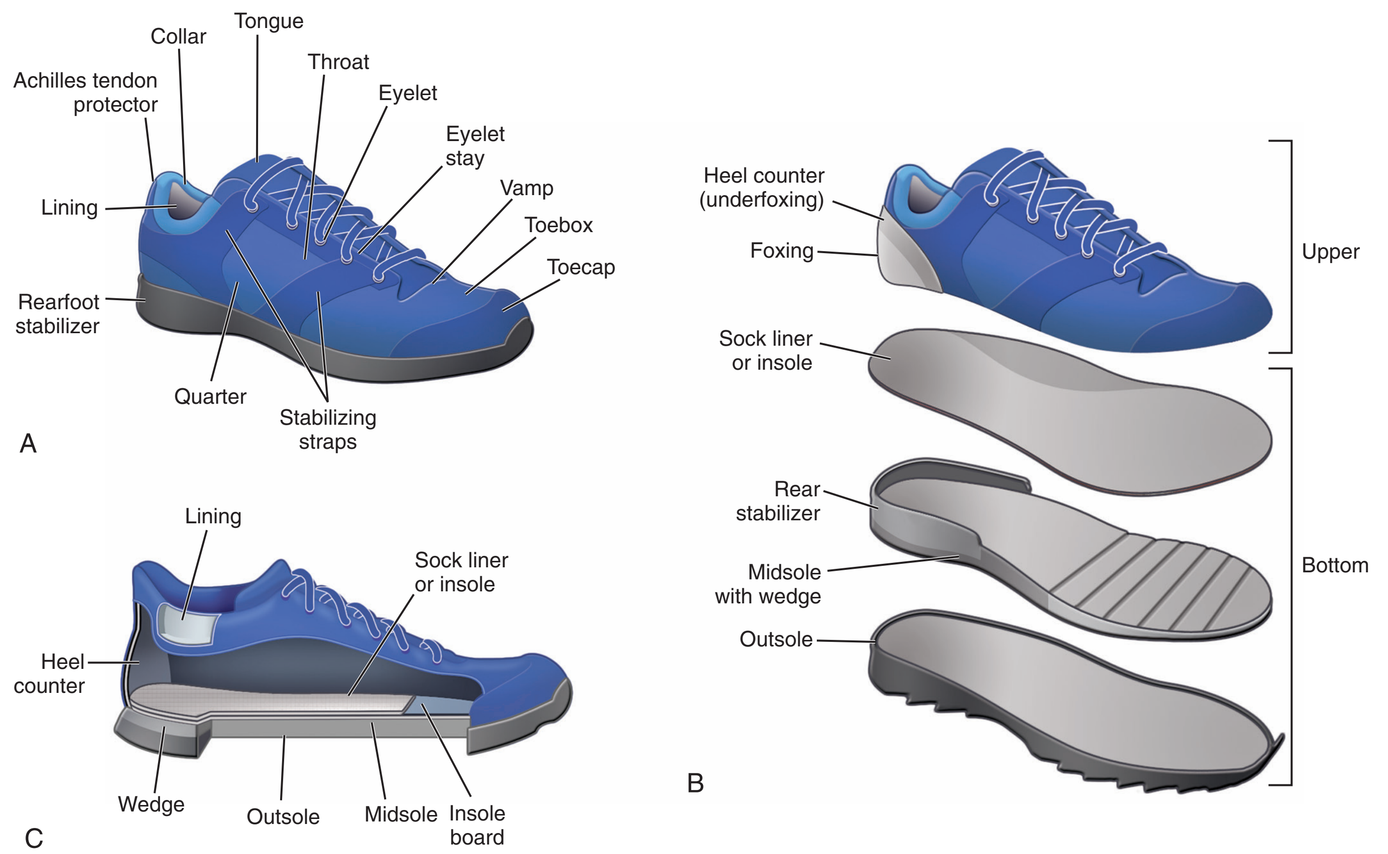


Fig. 116.4 Illustrations of athletic shoes. (A) Overview of external appearance. (B) Separation of shoe into component parts. (C) Sectional view of interior of shoe.

that the mechanical characteristics of midsole materials deteriorate significantly under normal use.[61] Thicker materials may deteriorate at a higher rate than thinner materials as well.[61] The rule of thumb is to replace athletic footwear after every 4 to 6 months of regular use.

The geometry of the midsole has also received much attention from shoe designers. Frequently incorporated into the modern athletic shoe midsole is sparse structure and open cantilever or honeycomb structures intended to vary mechanical characteristics and reduce weight.[63] These geometric features are intended to influence the compressibility, energy return, and flexibility of the midsole. Midsole geometry may also incorporate corrugations of various shapes, sipes, and other features to create directionally oriented flexion characteristics in the shoe sole.[64,65] More recently, some manufacturers have begun adding composite structures in the wedge or midsole to enhance mediolateral stability of the heel, midfoot, or forefoot of the shoe. One example of this approach is New Balance Rollbar Technology.

Any discussion of athletic shoe soles would be incomplete without mention of heel and forefoot rockers. There is considerable evidence to support the role of these features in influencing the kinematics and kinetics of the lower extremity. One recent study found that running with rocker soles may lower mechanical loading on the Achilles tendon; however, loading on the knee joint was found to be increased.[66]

Insole

The shoe insole connects the midsole to the foot. This layer is fabricated using resilient materials like polyurethanes. The geometry of the insole may also be designed to support the medial longitudinal arch. Arch support provided by the insole may vary depending on manufacturer and on the intended application of the shoe. For example, cross trainers may in general have less arch support than running shoes. Many athletic shoes are designed with removable insoles to simplify cleaning and accommodate custom foot orthoses. It should be noted, however, that the interface layer deep to the insole plays a significant role in the overall postural support of a custom arch support. In fact, the fit and function of flexible custom orthoses is typically adjusted by grinding the bottom surface of the insert. This fitting method alters the contour of the insert with respect to the shoe to influence the postural support of the foot.

Upper

The modern athletic shoe upper is fabricated by weaving, stitching, molding, and gluing operations. The shoe is assembled over a "last," a foot-shaped model that defines the interior shape of the shoe. Standards have been established to define the shapes of shoe lasts. More recently, some authors have proposed expanding these standards to better accommodate a broader range of foot shapes.[67] Manufacturers customize their lasts to meet the design specifications of their product line, sometimes employing the same last in the fabrication of different models of shoes.

The toe box, shoe collar, and heel counter protect the foot and provide postural support to the ankle. High-top uppers, for example, have been shown to decrease ankle inversion during cutting movements, and collar height may play a more significant role than counter stiffness in mediolateral stability.[68,69] More recently tactile elements have been developed that may find their way into the native shoe insole and upper with the goal of eliciting active arch elevation through the neuromotor response.[70]

High-top uppers are typically used for basketball shoes with designs that wrap around the malleoli but they are lower posteriorly to permit free plantarflexion.

Selection and Fitting of Sports Shoes

Some authors have recently proposed two new paradigms for running shoe selection, including the "preferred movement path" and the "comfort filter." The authors identify studies showing that shoes and inserts can significantly affect the incidence of injury, raising questions regarding previously accepted predictors of injury such as pronation and impact forces. On the basis of prior data indicating that shoes and/or shoe insoles did influence the range of skeletal movement, the authors propose that a "good running shoe" would be one that facilitates the individual runner's "preferred movement path." Furthermore, the authors acknowledge that shoe comfort is associated with a lower frequency of movement-related injury as well as lower oxygen consumption. Last, they propose that the runner automatically selects the shoe that facilitates his or her "preferred movement path" as the one that is most comfortable.[57]

The approach to orthotic treatment whereby the clinician considers both objective and subjective clinical indicators is consistent in this approach. The clinician familiar with the patient's history of orthotic use and fitting preferences can leverage the individual's awareness of shoe models and orthotic designs to help him or her find the most suitable athletic footwear and achieve a comfortable fit.

Perhaps those with greatest familiarity of sports shoe models and their features are the sales representatives at athletic shoe stores. These representatives may be skilled in helping their customers identify the subtle differences between sports shoe brands and models. The athlete should be encouraged to consult these specialty stores regarding options and features but cautioned that there is little objective evidence to associate sports shoe features with injury prevention.

When shoes are being fitted, a useful starting point to determine shoe size is measurement of foot length using the Brannock device.[72] This device was invented by Charles Brannock in 1927 and is the standard of measure for the world's footwear industry. Brannock devices are available at most shoe stores. It is important to note that although shoes are sized by the length of the longest toe on the longest foot, a more useful measure may be arch length. To measure the foot, place the patient's longest foot on the Brannock device and ask the patient to stand with the heel against the heel cup. Read the foot length by the longest toe. Then measure the arch length by aligning the arch length pointer with the first metatarsal head. To determine the shoe size, use the larger of the two measurements and add one shoe size to the measurement. It should be noted that this measure is only a starting point for shoe selection.

The American Orthopaedic Foot and Ankle Society (AOFAS) offers suggestions for selection of athletic footwear.[71] The AOFAS recommends purchasing athletic shoes from a specialty shoe store at the end of the day or after a workout when the feet are swollen. They recommend that shoes be sized while wearing the type of socks to be worn with the shoe, that there should be room for the toes and no slip of the heel, and finally that the shoe should feel comfortable at the first fitting while walking and running and during simulated activities that most closely approximate the sports activity.[71]

If the athlete participates in the sport three or more times per week, the AOFAS recommends purchase of a sports-specific shoe. It is important to note, however, that different manufacturers may build contradictory features into the same type of sports-specific shoe; therefore this selection criterion should be used with caution.

If there are specific pathologic musculoskeletal conditions, the clinician and athlete may find Table 116.3 a helpful guideline to aid in the selection of the best type of orthosis.

For a complete list of references, go to ExpertConsult.com.

TABLE 116.3 Guideline for the Selection of Sports Shoe Features

Pathology	Shoe Feature
Sesamoiditis with no other postural abnormality	• Cushioned insole • Rigid forefoot • Distal sole rocker
Metatarsalgia	• Cushioned insole • Rigid forefoot • Distal sole rocker
Morton neuroma with no other postural abnormality	• Cushioned insole • Rigid forefoot • Distal sole rocker
Plantar fasciitis with mild pes planus	• Cushioned heel • Supportive arch • Rigid forefoot • Distal sole rocker
Mild pes planus	• Supportive arch • Rigid forefoot • Distal sole rocker
Moderate to severe pes planus	• Supportive arch • Rigid forefoot • Distal sole rocker
Pes cavus	• Supportive arch • Rigid forefoot • Distal sole rocker • Surgical opening
Diabetes	• Extra depth • Wide last • Cushioned insole • Rigid forefoot • Distal sole rocker

SELECTED READINGS

Citation:
Guldemond NA, Leffers P, Schaper NC, et al. The effects of insole configurations on forefoot plantar pressure and walking convenience in diabetic patients with neuropathic feet. *Clinical Biomechanics*. 2007;22(1):81–87.

Level of Evidence:
II

Summary:
The effect of medial longitudinal arch pads, metatarsal pads, and full-foot wedges on the plantar pressure distribution of a foot orthosis were investigated in neuropathic patients. The effect of medial longitudinal and metatarsal pads appeared to be additive in decreasing pressure at the central and medial forefoot with less pressure relief at the lateral forefoot.

Citation:
Kogler GF, Solomonidis SE, Paul JP. Biomechanics of longitudinal arch support mechanisms in foot orthoses and their effect on plantar aponeurosis strain. *Clin Biomech (Bristol, Avon)*. 1996;11:243–252.

Level of Evidence:
II

Summary:
The effect of medial longitudinal arch supports on plantar fascial strain was investigated in cadaveric specimens. Foot orthoses were fabricated using foot molds created by elevating the medial longitudinal arch; these decreased plantar fascial strain more than prefabricated and functional foot orthoses.

Citation:
Mundermann A, Nigg BM, Humble RN, et al. Orthotic comfort is related to kinematics, kinetics, and EMG in recreational runners. *Med Sci Sports Exerc*. 2003;35(10):1710–1719.

Level of Evidence
II

Summary:
The effects of foot orthosis posting and molding on lower extremity kinematics, kinetics, electromyography, and perceived comfort were investigated in recreational runners. There appeared to be a significant correlation between the perceived comfort of foot orthoses and measured biomechanical variables.

Citation:
Mills K, Blanch P, Chapman AR, et al. Foot orthoses and gait: a systematic review and meta-analysis of literature pertaining to potential mechanisms. *Br J Sports Med*. 2010;44(14): 1035–1046.

Level of Evidence:
II

Summary:
A systematic review of the effects of foot orthosis posting and molding on lower extremity kinetics and kinematics in normal individuals. Foot orthosis posting reduced peak hindfoot eversion and tibial internal rotation, molding attenuated shock. The neuromotor influence of foot orthoses remained unclear.

REFERENCES

1. Schuster RO. A history of orthopedics in podiatry. *J Am Podiatry Assoc*. 1974;64(5):332–345.
2. Kay HW. A proposed nomenclature for limb prosthetics. *Orthotics and Prosthetics*. 1974;28(4):37–47.
3. Ball KA, Afheldt MJ. Evolution of foot orthotics – part 1: coherent theory or coherent practice? *J Manipulative Physiol Ther*. 2002;25(2):116–124.
4. Kirby KA. Evolution of foot orthoses in sports. In: Werd MB, Knight EL, eds. *Athletic Footwear and Orthoses in Sports Medicine*. New York: Springer; 2010:19–35.
5. Chevalier TL, Chockalingam N. Effects of foot orthoses: how important is the practitioner? *Gait Posture*. 2012;35(3):383–388.
6. Stefanyshyn DJ, Hettinga BA. Running injuries and orthotics. *International Sportmed Journal*. 2006;7(2):109–119.
7. Nicolopoulos CS, Scott BW, Giannoudis PV. Biomechanical basis of foot orthotic prescription. *Curr Orthop*. 2000;14(6): 464–469.
8. Hawke F, Burns J, Landorf KB. Evidence-based podiatric medicine importance of systematic reviews in clinical practice. *J Am Podiatr Med Assoc*. 2009;99(3):260–266.
9. Burns J, Landorf KB, Ryan MM, et al. Interventions for the prevention and treatment of pes cavus. *Cochrane Database Syst Rev*. 2007;(4):CD006154.
10. Shih YF, Wen YK, Chen WY. Application of wedged foot orthosis effectively reduces pain in runners with pronated foot: a randomized clinical study. *Clin Rehabil*. 2011;25(10):913–923.
11. Barton CJ, Munteanu SE, Menz HB, et al. The efficacy of foot orthoses in the treatment of individuals with patellofemoral pain syndrome. *Sports Medicine*. 2010;40(5):377–395.
12. Yeung SS, Yeung EW, Gillespie LD. Interventions for preventing lower limb soft-tissue running injuries. *Cochrane Database Syst Rev*. 2011;(7):CD001256.
13. Hirschmuller A, Baur H, Muller S, et al. Clinical effectiveness of customised sport shoe orthoses for overuse injuries in runners: a randomised controlled study. *Br J Sports Med*. 2011;45(12): 959–965.
14. McCormick F, Nwachukwu BU, Provencher MT. Stress fractures in runners. *Clin Sports Med*. 2012;31:291–306. United States.
15. Gross ML. Treatment of lower extremity injuries with orthotic shoe inserts. An overview. *Sports medicine (Auckland)*. 1993;15(1):66–70.
16. Gross ML. Effectiveness of orthotic shoe inserts in the long-distance runner. *Am J Sports Med*. 1991;19(4):409–412.
17. Hume P, Hopkins W, Rome K, et al. Effectiveness of foot orthoses for treatment and prevention of lower limb injuries – a review. *Sports Medicine*. 2008;38(9):759–779.
18. Kogler GF, Solomonidis SE, Paul JP. Biomechanics of longitudinal arch support mechanisms in foot orthoses and their effect on plantar aponeurosis strain. *Clin Biomech (Bristol, Avon)*. 1996;11:243–252.
19. Kitaoka HB, Luo ZP, An KN. Analysis of longitudinal arch supports in stabilizing the arch of the foot. *Clin Orthop Relat Res*. 1997;341:250–256.
20. Guldemond NA, Leffers P, Schaper NC, et al. The effects of insole configurations on forefoot plantar pressure and walking convenience in diabetic patients with neuropathic feet. *Clin Biomech (Bristol, Avon)*. 2007;22(1):81–87.
21. Baldassin V, Gomes CR, Beraldo PS. Effectiveness of prefabricated and customized foot orthoses made from low-cost foam for noncomplicated plantar fasciitis: a randomized controlled trial. *Arch Phys Med Rehabil*. 2009;90:701–706. United States.
22. Landorf KB, Keenan AM, Herbert RD. Effectiveness of foot orthoses to treat plantar fasciitis – a randomized trial. *Arch Intern Med*. 2006;166(12):1305–1310.
23. Brodtkorb TH, Kogler GF, Arndt A. The influence of metatarsal support height and longitudinal axis position on plantar foot loading. *Clin Biomech (Bristol, Avon)*. 2008;23:640–647. England.
24. Chiu HT, Shiang TY. Effects of insoles and additional shock absorption foam on the cushioning properties of sport shoes. *J Appl Biomech*. 2007;23(2):119–127.
25. Perhamre S, Lundin F, Klassbo M, et al. A heel cup improves the function of the heel pad in Sever's injury: effects on heel pad thickness, peak pressure and pain. *Scandinavian journal of medicine & science in sports*. 2012;22(4):516–522.
26. Huerta JP, Moreno JMR, Kirby KA, et al. Effect of 7-degree rearfoot varus and valgus wedging on rearfoot kinematics and kinetics during the stance phase of walking. *J Am Podiatr Med Assoc*. 2009;99(5):415–421.
27. Huerta JP, Moreno JMR, Kirby KA. Static response of maximally pronated and nonmaximally pronated feet to frontal plane wedging of foot orthoses. *J Am Podiatr Med Assoc*. 2009;99(1): 13–19.
28. Tsung BYS, Zhang M, Mak AFT, et al. Effectiveness of insoles on plantar pressure redistribution. *J Rehabil Res Dev*. 2004;41(6A):767–774.
29. Kogler GF, Solomonidis SE, Paul JP. In vitro method for quantifying the effectiveness of the longitudinal arch support mechanism of a foot orthosis. *Clin Biomech (Bristol, Avon)*. 1995;10:245–252.
30. Kogler GF, Veer FB, Solomonidis SE, et al. The influence of medial and lateral placement of orthotic wedges on loading of the plantar aponeurosis. *J Bone Joint Surg Am*. 1999;81(10): 1403–1413.
31. Mundermann A, Nigg BM, Humble RN. Stefanyshyn DJ. Orthotic comfort is related to kinematics, kinetics, and EMG in recreational runners. *Med Sci Sports Exerc*. 2003;35(10): 1710–1719.
32. Nurse MA, Hulliger M, Wakeling JM, et al. Changing the texture of footwear can alter gait patterns. *J Electromyogr Kinesiol*. 2005;15(5):496–506.
33. Nigg BM, Nurse MA. Stefanyshyn DJ. Shoe inserts and orthotics for sport and physical activities. *Med Sci Sports Exerc*. 1999;31(7 suppl):S421–S428.
34. Mills K, Blanch P, Vicenzino B. Comfort and midfoot mobility rather than orthosis hardness or contouring influence their immediate effects on lower limb function in patients with anterior knee pain. *Clin Biomech (Bristol, Avon)*. 2012;27:202–208. England: 2011 Elsevier Ltd.
35. Ritchie C, Paterson K, Bryant AL, et al. The effects of enhanced plantar sensory feedback and foot orthoses on midfoot kinematics and lower leg neuromuscular activation. *Gait & Posture*. 2011;33(4):576–581.
36. Mundermann A, Nigg BM, Humble RN, et al. Foot orthotics affect lower extremity kinematics and kinetics during running. *Clin Biomech (Bristol, Avon)*. 2003;18:254–262. England.
37. Nigg BM, Stergiou P, Cole G, et al. Effect of shoe inserts on kinematics, center of pressure, and leg joint moments during running. *Med Sci Sports Exerc*. 2003;35(2):314–319.
38. van Gent RN, Siem D, van Middelkoop M, et al. Incidence and determinants of lower extremity running injuries in long

distance runners: a systematic review. *Br J Sports Med*. 2007;41(8):469–480.
39. Wilder RP, Sethi S. Overuse injuries: tendinopathies, stress fractures, compartment syndrome, and shin splints. *Clin Sports Med*. 2004;23(1):55–81.
40. Di Caprio F, Buda R, Mosca M, et al. Foot and lower limb diseases in runners: assessment of risk factors. *Journal of Sports Science and Medicine*. 2010;9(4):587–596.
41. Chandler TJ, Kibler WB. A biomechanical approach to the prevention, treatment and rehabilitation of plantar fasciitis. *Sports Med*. 1993;15(5):344–352.
42. Fields KB, Sykes JC, Walker KM, et al. Prevention of running injuries. *Curr Sports Med Rep*. 2010;9:176–182. United States.
43. Hockenbury RT. Forefoot problems in athletes. *Med Sci Sports Exerc*. 1999;31(7):S448–S458.
44. Imhauser CW, Abidi NA, Frankel DZ, et al. Biomechanical evaluation of the efficacy of external stabilizers in the conservative treatment of acquired flatfoot deformity. *Foot & Ankle International*. 2002;23(8):727–737.
45. Hawke F, Burns J, Radford JA, et al. Custom-made foot orthoses for the treatment of foot pain. *Cochrane Database Syst Rev*. 2008;(3):CD006801, 135.
46. Mills K, Blanch P, Vicenzino B. Influence of contouring and hardness of foot orthoses on ratings of perceived comfort. *Med Sci Sports Exerc*. 2011;43(8):1507–1512.
47. Sedov AV, Vazhbin LB, Odtarzhevskaia ND, et al. Use of insoles made of antimicrobial materials as prophylactic means in foot mycoses. *Med Tr Prom Ekol*. 1994;7:29–32.
48. Lu XF. *Measurements of the Friction Coefficient of Insole Fabric. Textile Bioengineering and Informatics Symposium Proceedings*. Vol. 1-3. 2010:1390–1394.
49. Paton J, Jones RB, Stenhouse E, et al. The physical characteristics of materials used in the manufacture of orthoses for patients with diabetes. *Foot & Ankle International*. 2007;28(10):1057–1063.
50. Mills K, Blanch P, Chapman AR, et al. Foot orthoses and gait: a systematic review and meta-analysis of literature pertaining to potential mechanisms. *Br J Sports Med*. 2010;44(14):1035–1046.
51. Bus SA, Ulbrecht JS, Cavanagh PR. Pressure relief and load redistribution by custom-made insoles in diabetic patients with neuropathy and foot deformity. *Clin Biomech (Bristol, Avon)*. 2004;19(6):629–638.
52. Jong E. Shoes – fashion and fantasy – McDowell, C. In: *New York Times Book Review*. 1989. 45–45.
53. Stewart SF. Footgear–its history, Uses and abuses. *Clin Orthop Relat Res*. 1972;88:119–130.
54. Cavanagh P. *The Running Shoe Book*. Mountain View, CA: Anderson World; 1980.
55. Wilcox R. *The Mode in Footwear*. New York: Charles Scribner and Sons; 1948.
56. Nigg BM, Vienneau J, Smith AC, et al. The preferred movement path paradigm: influence of running shoes on joint movement. *Med Sci Sports Exerc*. 2017.
57. Nigg BM, Baltich J, Hoerzer S, et al. Running shoes and running injuries: mythbusting and a proposal for two new paradigms: 'preferred movement path' and 'comfort filter'. *Br J Sports Med*. 2015;49(20):1290–1294.
58. Harada M, Kumagawa H, Izumi J, et al. *Shoe for Sports Involving Running*. USA: 1990.
59. Derler S, Kausch F, Huber R. Analysis of factors influencing the friction coefficients of shoe sole materials. *Safety Science*. 2008;46(5):822–832.
60. Blanchette MG, Powers CM. The influence of footwear tread groove parameters on available friction. *Appl Ergon*. 2015;50:237–241.
61. Lippa NM, Collins PK, Bonacci J, et al. Mechanical ageing performance of minimalist and traditional footwear foams. *Footwear Science*. 2017;9(1):9–20.
62. Norton D, Adidas International Marketing B.V., assignee. *Sole Elements of Varying Density and Methods of Manufacture*. USA; 2008.
63. Grote P, Choe P, Murphy S, et al, inventor; New Balance Athletic Shoe, Inc., assignee. Mechanical Cushioning System for Footwear. US patent application US20100126040A1. May 27, 2010.
64. Cin D, inventor; Nike, Inc., assignee. Sole Structures and Articles of Footwear Having an Elongated Hexagonal Siping Pattern and/or a Heel Pocket Structure. US patent application US20170099907A1. April 13, 2017.
65. Bunnell DD, Farris BN, Sheets-Singer A, et al., inventors; Nike, Inc., assignee. Sole Structure for an Article of Footwear Having a Nonlinear Bending Stiffness with Compression Grooves and Descending Ribs. US patent application US20170127755A1. May 11, 2017.
66. Sobhani S, van den Heuvel ER, Dekker R, et al. Biomechanics of running with rocker shoes. *J Sci Med Sport*. 2017;20(1):38–44.
67. Nakayama K, Abe K, Nagai K, et al. Establishing reference sizes for the rear portion of the foot for use in designing lasts for women's shoes. *Footwear Science*. 2017;9(1):33–39.
68. Bishop JL, Nurse MA, Bey MJ. Do high-top shoes reduce ankle inversion? A dynamic x-ray analysis of aggressive cutting in a high-top and low-top shoe. *Footwear Science*. 2014;6(1):21–26.
69. Liu H, Wu Z, Lam WK. Collar height and heel counter-stiffness for ankle stability and athletic performance in basketball. *Res Sports Med*. 2017;25(2):209–218.
70. Kohatsu SS, Cook CS, Schoolmeester B, et al., inventors; Nike Innovate C.V., assignee. Article of Footwear with a Tactile Feedback System. US patent WO/2017/083195. April 11, 2016.
71. AOFAS. How to Select the Right Athletic Shoes. American Orthopaedic Foot and Ankle Society. AOFAS FootCare MD(R) website. http://www.aofas.org/footcaremd/how-to/footwear/pages/selecting-athletic-shoes.aspx. Accessed September 22, 2018.
72. The Brannock Device Company. Instructions & Fitting Tips. Genuine Brannock Device (R) website. https://brannock.com/pages/instructions-fitting-tips. Accessed September 22, 2018.
73. Deleted in review.
74. Deleted in review.
75. Deleted in review.
76. Deleted in review.
77. Yavuz M, Tajaddini A, Botek G, et al. Temporal characteristics of plantar shear distribution: relevance to diabetic patients. *J Biomech*. 2008;41(3):556–559.
78. Leitch KM, Birmingham TB, Jones IC, et al. In-shoe plantar pressure measurements for patients with knee osteoarthritis: reliability and effects of lateral heel wedges. *Gait & Posture*. 2011;34(3):391–396.
79. Bonanno DR, Zhang CY, Farrugia RC, et al. The effect of different depths of medial heel skive on plantar pressures. *Journal of Foot and Ankle Research*. 2012;5.
80. Shelburne KB, Torry MR, Steadman JR, et al. Effects of foot orthoses and valgus bracing on the knee adduction moment and medial joint load during gait. *Clin Biomech (Bristol, Avon)*. 2008;23:814–821.